

Multi-Cloud AI Infrastructure Market Analysis

Comparative Analysis of AWS, Azure, and Google Cloud Platform

Author

Che-Wei Lee

Program

Master of Science in Data Analytics Engineering

Institution

Northeastern University

Date

June 2026

1. Executive Summary

This project analyzes the cloud infrastructure and AI ecosystems of AWS, Microsoft Azure, and Google Cloud Platform (GCP). Using publicly available cloud service, regional deployment, and AI accelerator data, the study compares provider capabilities through interactive Tableau dashboards and business intelligence analysis.

The results show that AWS maintains the largest service ecosystem, Azure operates the broadest global infrastructure footprint, and GCP focuses heavily on AI innovation and accelerator technologies. Overall, the findings suggest that future cloud competition will be increasingly shaped by AI services, specialized hardware, and global infrastructure expansion.

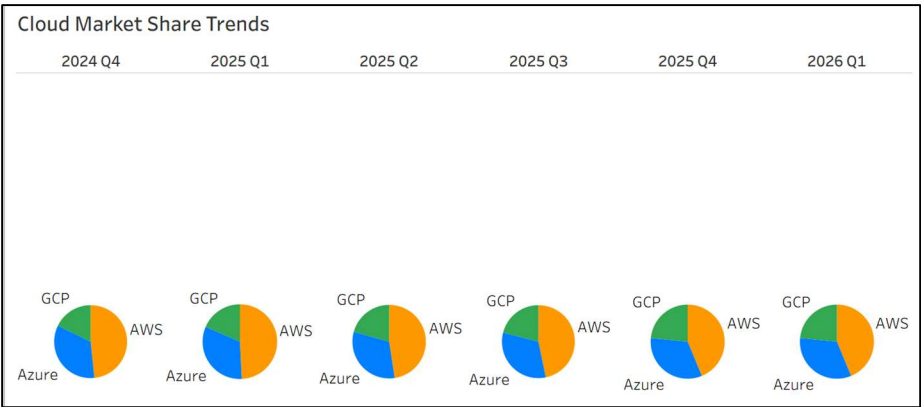
2. Project Background

2.1 Overview of the Cloud Computing Market

Cloud computing has fundamentally changed how organizations access and manage IT resources. Instead of maintaining physical infrastructure, businesses can leverage on-demand computing, storage, networking, and software services through cloud platforms.

Today, the global cloud market is dominated by three major providers: Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Together, they offer a wide range of services that support enterprise operations, data analytics, artificial intelligence, and digital transformation initiatives.

As cloud adoption continues to grow, competition among providers is increasingly driven by AI capabilities, infrastructure scale, and service innovation.

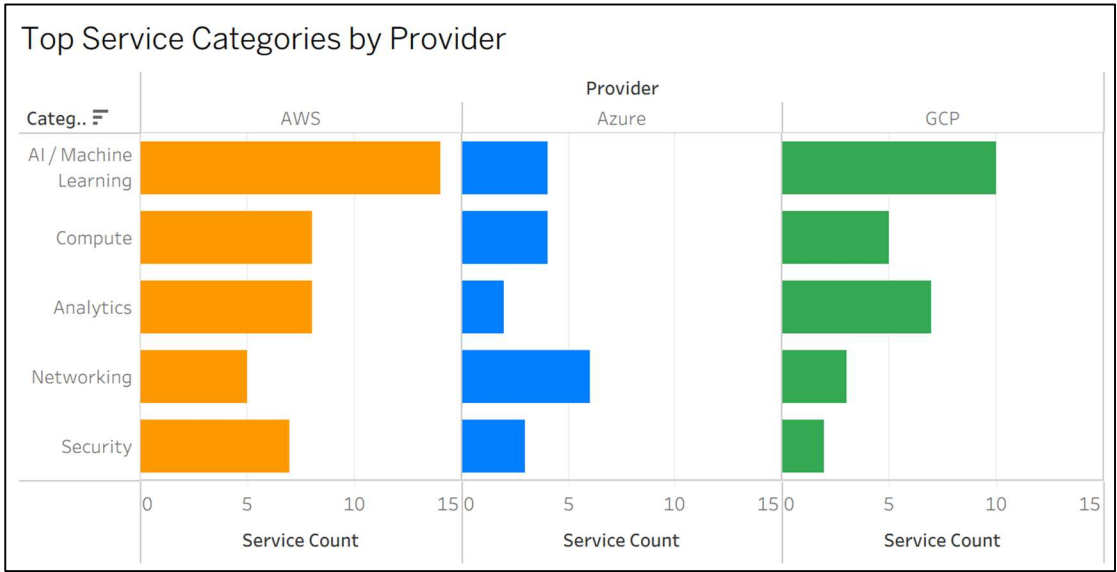


2.2 Growth of Artificial Intelligence Infrastructure

The rapid advancement of artificial intelligence technologies has significantly increased demand for cloud-based computing resources. Large language models, generative AI applications, machine learning platforms, and data-intensive analytics workloads require substantial computational power and scalable infrastructure.

To support these requirements, cloud providers have expanded their investments in GPU-enabled virtual machines, AI development platforms, managed machine learning services, and specialized hardware accelerators.

As AI adoption becomes a strategic priority for businesses worldwide, cloud providers increasingly compete through AI service offerings, infrastructure capacity, and ecosystem development.



2.3 Research Motivation

The cloud computing industry is evolving rapidly, particularly in response to growing demand for artificial intelligence applications. While AWS, Azure, and GCP all provide extensive cloud ecosystems, their service portfolios, AI offerings, GPU infrastructure, and regional deployment strategies differ significantly.

Understanding these differences is valuable for organizations seeking cloud solutions, researchers studying cloud market dynamics, and technology professionals evaluating infrastructure trends.

This project was developed to provide a data-driven comparison of the three major cloud providers by integrating multiple public datasets and visualizing key aspects of the AI cloud ecosystem. The project aims to identify competitive strengths, market positioning, and infrastructure characteristics across leading cloud platforms.

2.4 Project Scope

This study focuses on the following areas:

- Cloud service ecosystem analysis
- AI-related service availability
- GPU instance offerings and pricing
- Global cloud region distribution
- Marketplace service ecosystems
- Interactive business intelligence dashboards

The analysis is limited to publicly available information from AWS, Azure, and GCP and does not include proprietary enterprise pricing agreements or private infrastructure deployments.

3. Project Objectives

The primary goal of this project is to analyze and compare the AI infrastructure ecosystems of the three leading cloud providers: Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). By integrating cloud service catalogs, marketplace offerings, regional deployment information, GPU resources, and pricing data, this study aims to provide a comprehensive view of the current cloud computing landscape.

To achieve this goal, the project focuses on the following objectives:

1. Compare Cloud Service Ecosystems

Evaluate and compare the breadth and composition of cloud service portfolios across AWS, Azure, and GCP. The analysis examines service categories such as compute, storage, databases, networking, security, analytics, and artificial intelligence.

2. Analyze AI Service Availability

Identify and compare AI-related services offered by each cloud provider. This includes managed machine learning platforms, generative AI services, data science tools, and AI infrastructure solutions.

3. Evaluate GPU Infrastructure and Pricing

Analyze the availability, specifications, and pricing of GPU-enabled cloud resources. The objective is to understand how cloud providers support AI and high-performance computing workloads through specialized hardware offerings.

4. Examine Global Infrastructure Distribution

Investigate the geographical distribution of cloud regions and infrastructure deployment strategies. This analysis provides insight into global service coverage and regional expansion patterns.

5. Develop Interactive Business Intelligence Dashboards

Create interactive Tableau dashboards that enable users to explore cloud market trends, compare service ecosystems, and generate actionable insights from the collected datasets.

Expected Outcomes:

- Provide a structured comparison of AWS, Azure, and GCP service ecosystems.
- Identify major trends in AI-related cloud services.
- Reveal differences in GPU infrastructure availability and pricing strategies.
- Highlight regional deployment patterns across global cloud markets.
- Deliver interactive visualizations that support cloud market analysis and business decision-making.

4. Data Collection and Preparation

4.1 Data Collection

This project was developed by collecting and integrating public cloud infrastructure data from Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Multiple datasets were gathered to capture different aspects of the cloud market, including service catalogs, AI-related offerings, GPU resources, regional deployments, and pricing information.

The collected data provides a comprehensive view of cloud provider capabilities and enables cross-provider comparison of service ecosystems and AI infrastructure.

- Key Data Sources
- Cloud Service Catalogs
- AI Service Inventories
- GPU Resource Datasets
- Cloud Region Information
- Virtual Machine Pricing Data
- Marketplace Service Catalogs

4.2 Data Cleaning and Standardization

Since each cloud provider uses different naming conventions, service classifications, and regional structures, data cleaning and standardization were required before analysis.

The preparation process included:

- Removing duplicate records
- Handling missing values
- Standardizing provider names
- Normalizing service categories
- Aligning regional information
- Structuring pricing data for comparison

To support cross-provider analysis, services were grouped into common categories such as Compute, Storage, Database, Analytics, Security, Networking, and AI/ML.

4.3 Feature Engineering

Several analytical features were created to support business intelligence reporting and dashboard development.

These features included:

- AI-related service identification
- GPU resource classification
- Provider service category counts
- Regional infrastructure summaries
- AI service share calculations
- Pricing benchmark summaries

The transformed datasets enabled consistent comparison across AWS, Azure, and GCP.

4.4 Analytics Workflow

The project followed a structured analytics workflow:

1. Collect cloud service and infrastructure data from public sources.
2. Clean and standardize provider-specific datasets.
3. Create summary tables and analytical features.
4. Integrate datasets into a unified analytical model.
5. Develop interactive Tableau dashboards.
6. Generate insights on cloud ecosystems, AI services, GPU infrastructure, and regional deployment strategies.

4.5

Category	Tools
Data Collection	Public APIs, Cloud Catalog Data
Data Processing	Python, Pandas
Data Storage	CSV Files
Data Visualization	Tableau
Data Validation	Microsoft Excel

5. Dashboard Analysis

This chapter presents the exploratory analysis of the cloud infrastructure market across AWS, Azure, and Google Cloud Platform (GCP). The objective is to identify major patterns in cloud service ecosystems, AI-related offerings, infrastructure deployment, and GPU resource availability.

The analysis was conducted using the prepared datasets and visualized through interactive Tableau dashboards.

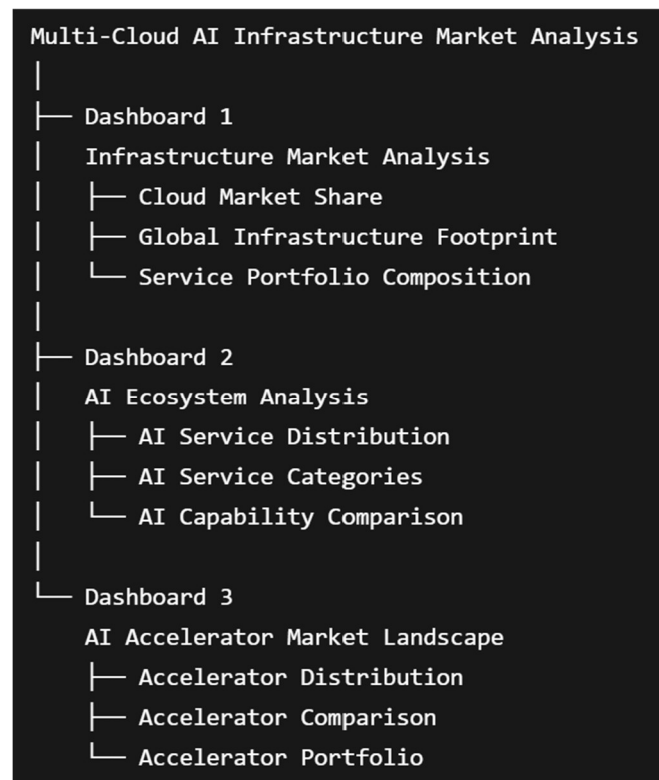
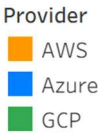


Figure 5 illustrates the overall analytical structure of the project. To evaluate the competitive landscape of AWS, Azure, and GCP, three Tableau dashboards were developed, each focusing on a different aspect of the cloud and AI infrastructure market.

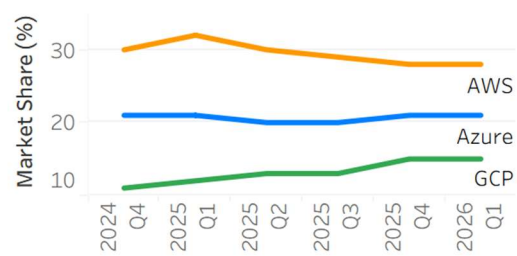
5.1 Dashboard 1: Multi-Cloud Infrastructure Market Analysis

Multi-Cloud Infrastructure Market Analysis

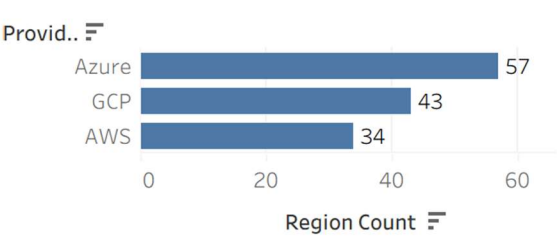
AWS remains the market leader (28%), Azure maintains the largest global footprint (57 regions), while GCP continues to gain market share through rapid AI-focused expansion.



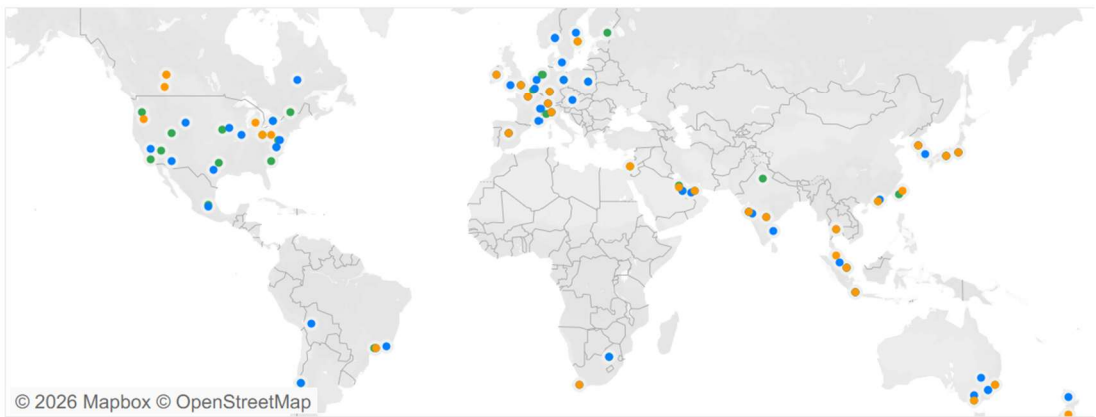
Cloud Market Share Trends



Regional Footprint Comparison



Global Cloud Infrastructure Footprint



Overview

This dashboard provides a high-level overview of the global cloud computing market by comparing market share trends, regional infrastructure footprints, and global deployment patterns across AWS, Azure, and Google Cloud Platform (GCP).

Key Insights

- AWS remains the market leader with the largest cloud market share and a globally diversified infrastructure network.
- Azure maintains the broadest regional footprint, operating across 57 cloud regions

worldwide.

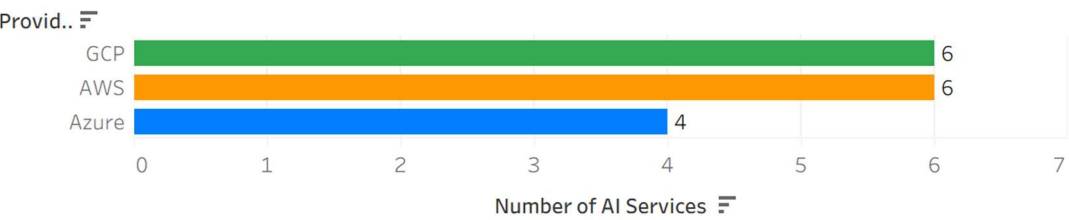
- GCP continues to expand its presence through AI-focused investments and strategic regional growth.
- North America, Europe, and Asia-Pacific remain the primary hubs of cloud infrastructure deployment.

5.2 AI Ecosystem Analysis

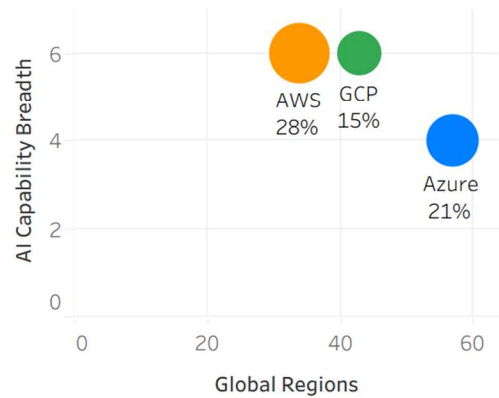
AI Ecosystem Analysis

AWS leads in market share and AI portfolio depth, Azure offers the broadest global infrastructure footprint, while GCP shows strong specialization across emerging AI capabilities.

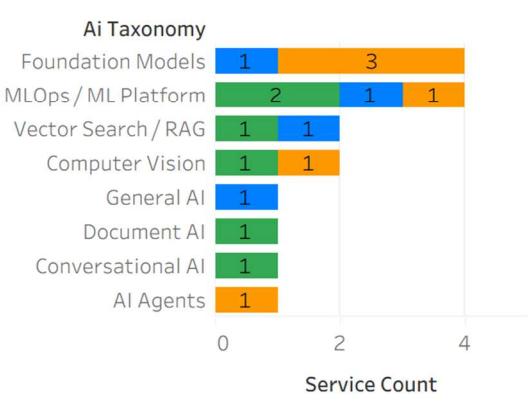
AI Service Portfolio Comparison



AI Competitive Positioning Matrix



AI Capability Distribution



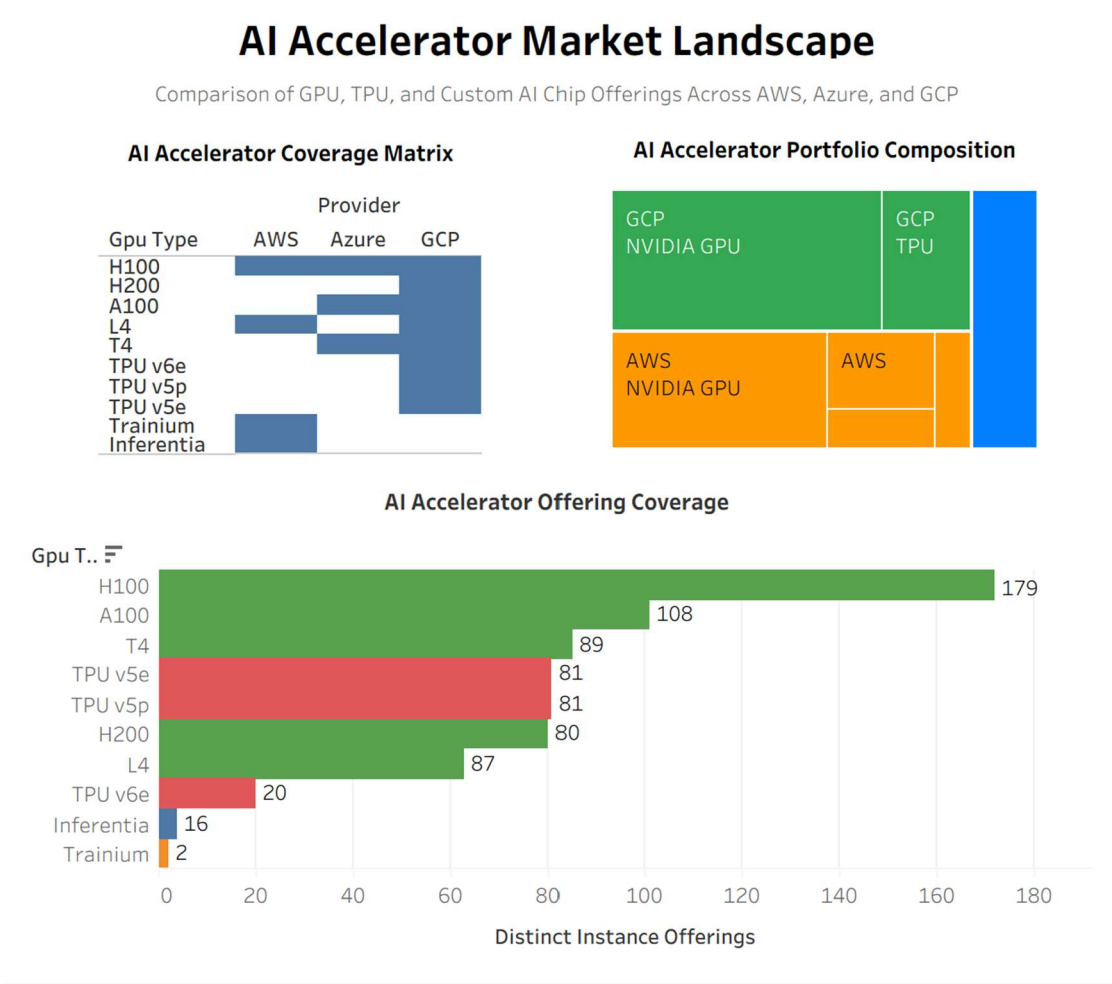
Overview

This dashboard evaluates the AI service ecosystems of AWS, Azure, and GCP by comparing AI service portfolios, capability breadth, and specialization across emerging AI domains.

Key Insights

- AWS and GCP offer the largest AI service portfolios among the three providers.
- Azure demonstrates strong infrastructure coverage but a relatively narrower AI service portfolio.
- Foundation Models, MLOps platforms, and Retrieval-Augmented Generation (RAG) capabilities have emerged as core components of modern AI ecosystems.
- AI has become a primary competitive differentiator in the cloud market.

5.3 AI Accelerator Market Landscape



Overview

This dashboard analyzes AI accelerator offerings across AWS, Azure, and GCP, including NVIDIA GPUs, Google TPUs, and provider-specific AI chips designed for training and inference

workloads.

Key Insights

- NVIDIA GPUs remain the dominant AI accelerator across all cloud providers.
- GCP differentiates itself through proprietary TPU offerings, including TPU v5e, TPU v5p, and TPU v6e.
- AWS expands its AI infrastructure through custom silicon solutions such as Trainium and Inferentia.
- H100 and A100 GPUs continue to be the most widely available accelerators for enterprise AI workloads.

5.4 Key Findings and Strategic Insights

Based on the three dashboards, several major trends emerge across the cloud infrastructure market:

- **Market Leadership**

AWS maintains its position as the overall market leader through its extensive service ecosystem, strong market share, and broad infrastructure coverage.

- **AI-Centric Competition**

Cloud competition is increasingly shifting from traditional infrastructure services toward AI platforms, foundation models, and AI development ecosystems.

- **Specialized Hardware Differentiation**

While NVIDIA GPUs remain the industry standard, cloud providers are investing heavily in proprietary AI accelerators to reduce costs and improve performance for large-scale AI workloads.

- **Global Infrastructure Expansion**

Cloud infrastructure remains concentrated in North America, Europe, and Asia-Pacific, reflecting the regions with the highest demand for enterprise cloud services and AI

computing resources.

6. Business Implications

The findings from this project highlight several important trends shaping the future of cloud computing and artificial intelligence infrastructure. As cloud providers continue expanding their service ecosystems and AI capabilities, competitive advantages are increasingly driven by platform breadth, infrastructure scale, and specialized AI technologies.

6.1 AI Has Become a Core Growth Driver

The analysis indicates that AI-related services are now a central component of cloud provider strategies. Beyond traditional computing and storage services, cloud platforms are increasingly competing through foundation models, machine learning platforms, AI development tools, and intelligent automation services.

Organizations adopting cloud technologies are therefore evaluating providers not only on infrastructure capabilities but also on the maturity of their AI ecosystems.

6.2 Infrastructure Scale Remains a Competitive Advantage

Regional coverage remains a critical factor in cloud adoption. Providers with broader infrastructure networks can offer lower latency, higher availability, and stronger regulatory compliance across global markets.

The analysis shows that AWS and Azure maintain extensive global footprints, enabling them to support enterprise customers across multiple geographic regions.

6.3 AI Accelerators Are Strategic Assets

The growing demand for generative AI and large language models has significantly increased the importance of specialized computing hardware.

While NVIDIA GPUs remain the dominant industry standard, cloud providers are increasingly investing in proprietary accelerator technologies such as Google's TPU family and AWS

Trainium and Inferentia chips. These investments may reduce infrastructure costs while improving performance for large-scale AI workloads.

6.4 Increasing Market Differentiation

Although AWS, Azure, and GCP all provide comprehensive cloud platforms, their strategic priorities differ.

- AWS emphasizes service breadth and ecosystem maturity.
- Azure leverages enterprise integration and global infrastructure coverage.
- GCP focuses on AI innovation, advanced analytics, and proprietary accelerator technologies.

These differences create distinct value propositions for organizations selecting cloud providers.

6.5 Implications for Future Cloud Adoption

As AI adoption continues to accelerate, cloud provider competition is expected to shift further toward AI infrastructure, specialized hardware, and intelligent service ecosystems. Organizations evaluating cloud strategies should consider not only current infrastructure capabilities but also each provider's long-term investment in AI technologies and global infrastructure expansion.

7. Limitations

While this project provides valuable insights into the cloud infrastructure and AI ecosystem landscape, several limitations should be considered.

Public Data Availability

The analysis relies on publicly available cloud service catalogs, pricing information, and

infrastructure data. Proprietary enterprise agreements, private pricing structures, and internal operational metrics were not accessible and therefore are not reflected in the results.

Dynamic Cloud Market

The cloud computing industry evolves rapidly. New services, AI capabilities, accelerator offerings, and regional expansions are introduced frequently, meaning that the findings represent a snapshot of the market at the time of data collection.

Service Classification Assumptions

To enable cross-provider comparison, services were grouped into standardized categories. Although this improves consistency, some services may overlap multiple categories and require subjective classification decisions.

Limited Pricing Scope

Pricing comparisons were based on publicly available information and selected infrastructure offerings. Actual costs may vary depending on region, workload requirements, reserved pricing models, and enterprise contracts.

Scope of Providers

This project focuses exclusively on AWS, Microsoft Azure, and Google Cloud Platform. Other cloud providers such as Oracle Cloud Infrastructure (OCI), Alibaba Cloud, and IBM Cloud were not included in the analysis.

Future Enhancement Opportunities

Future versions of this project could incorporate:

- Additional cloud providers
- Real-time pricing pipelines
- Automated data collection workflows
- GPU performance benchmarking
- Longitudinal market trend analysis
- Cloud adoption forecasting models

8. Conclusion

This project presented a comprehensive analysis of the cloud infrastructure market across Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). By integrating cloud service catalogs, AI service portfolios, regional infrastructure data, and AI accelerator offerings, the study provided a data-driven comparison of the three leading cloud providers.

The analysis revealed that AWS continues to maintain its leadership position through its extensive service ecosystem and strong market presence. Azure demonstrated the broadest global infrastructure footprint, highlighting its commitment to enterprise-scale cloud adoption and regional accessibility. Meanwhile, GCP differentiated itself through its strong focus on artificial intelligence, advanced analytics, and proprietary accelerator technologies.

The AI ecosystem analysis showed that artificial intelligence has become a major competitive driver within the cloud industry. Foundation models, MLOps platforms, retrieval-augmented generation (RAG) solutions, and AI development services are increasingly shaping cloud provider strategies and influencing customer adoption decisions.

The AI accelerator analysis further highlighted the growing importance of specialized computing hardware. While NVIDIA GPUs remain the dominant accelerator technology across the industry, cloud providers are actively investing in proprietary solutions such as Google's TPU family and AWS Trainium and Inferentia chips to support next-generation AI workloads.

Overall, the findings suggest that competition within the cloud market is evolving beyond traditional infrastructure services and increasingly centered on AI capabilities, specialized hardware, and ecosystem development. As organizations continue to adopt AI-driven technologies, cloud providers that successfully combine scalable infrastructure, advanced AI services, and innovative accelerator technologies will be best positioned to maintain long-term competitive advantages.

This project demonstrates how data integration, visualization, and business intelligence techniques can be applied to transform complex cloud market data into actionable insights. The resulting dashboards and analytical framework provide a foundation for future research into cloud computing, AI infrastructure, and emerging technology markets.